PHS 2000B Problem Set 7: Regression discontinuity

Due April 9, 2026 at 11:59pm

We encourage students to help each other out on problem sets. While the final product that you turn in must be your own individual work that you have written up in isolation from your partners, you can still seek the help of your peers if you get stuck on a particular question. We request that you write the names of your collaborators below.

Collaborators:

Part 1: True, false, or uncertain

Are the following statements True (T), False (F), or Uncertain (U). If true, state in a full sentence why this statement is true. If false, explain why it is false and when possible give a specific example of a case where the statement will not be true. You should respond (U) if you believe the statement can be either T or F depending on what assumptions you make and you should give examples of both possibilities.

1) In a sharp RDD, the average treatment effect at the threshold is the average treatment effect. (1 pt)

Solutions

False, because a sharp RDD exploits a discontinuity at the threshold, our treatment effect of interest is focused at the threshold. In RDD, we refer to this as the average causal effect, to avoid confusion. To assume that the ACE = ATE, we would need to assume that the average treatment effect is homogenous across the running variable, which is a large assumption.

2) Imagine you are using a regression discontinuity design to identify the impact of a program that pays down utility bills for low-income households. In your proposed design, household income is your running variable and the threshold for program eligibility is 200% of the federal poverty line. In some states, 200% of the federal poverty line is also the threshold for eligibility for Medicaid. Claim: Your research design is able to isolate the causal effect of the utility program with a fuzzy RDD design. (1 pt)

Solution

False. Access to multiple different types of programs could violate our continuity assumption. If we violate the continuous potential outcomes assumption, we can neither validly identify the causal effect in a fuzzy RDD nor a sharp one.

3) RDD estimates don't tell us much about how the treatment would impact those with values of the running variable that are far away from the threshold. (1 pt)

Solution

True. The RDD estimate as the causal effect at the threshold which may not be the same effect for those with values of the running variable far away from the threshold

Part 2: RDD Assumptions

4) You have been asked to review a paper which proposes a sharp RD design. For each statement below, indicate how the statement would affect your assessment of the internal validity of the RDD estimates. Limit your explanation to 1-2 sentences per statement. (5 pts)

a.Individuals in the study do not know the value of the running variable.

b.Individuals know and self-report the value of the running variable that determines program eligibility. Assume that the running variable is not fixed and has the potential to change over time, that the self-report is not independently verified, and people know that they will be taken at their word.

c.Individuals know and self-report the value of the running variable that determines eligibility for a program, but the running variable itself is a fixed characteristic. Assume that the self-report is not independently verified, and people know that they will be taken at their word.

d.Values of the running variable are measured by a process that is entirely independent of the application of treatment.

e.Individuals are aware of the cut-off and appear to factor the cut-off into their behavior.

Solution

a. This is a desirable characteristic of the running variable, as it makes sorting less plausible

b. This would raise the possibility of sorting, as people may self-report values mindful of the threshold.

c. Again, we have an elevated possibility of sorting, as people may self-report values mindful of the threshold. If the self-report is not verified, a fixed characteristic can be just as subject to sorting as a more flexible one if individuals know that their self-report is not checked.

d. This is encouraging and should lessen our concern about sorting.

e. This indicates that there is clear sorting around the threshold.

5) Choose two of the above statements and provide an example of a running variable that is qualitatively similar to what is described in the statement. (2 pts)

Solutions

Answers may vary. A list of examples for each below:

a. Birth weight is a running variable in which the birthing person does not know the exact number until the moment the baby is weighed at delivery. CD4 count is another measure that the individual themself wouldn't know until measured during study/ healthcare procedures.

b.Certain income-based programs (such as some free lunch programs in public schools)

c. Age cutoffs for discounts at restaurants. Age (or at least date of birth) is fixed, but that does not mean the individual cannot mis-report. *Tough to think of good one for this statement*

d.SAT is administered by test proctors, while the decision to award a scholarship based on SAT scores is made by university financial aid officers

e.Some people strategically delay earning income in a given tax year to qualify for a tax credit that they would otherwise not be eligible for.

Part 3: Short answer

The Ministry of Health and Environment of a North American city implements an air-quality alert program. The program works as follows: every day, the Ministry forecasts a measure known as the Air Quality Index (AQI). The AQI is a function of the forecasted concentrations of different pollutants in the city during that day. If the AQI is greater than or equal to 48, then an alert is sent to all residents of the city warning them about poor air quality. You understand that when AQI is greater than or equal to 48, an alert is always issued without fail; when AQI<48, an alert is never issued.

Let

X**: AQI**

Y: **cardiovascular–related mortality**

A: **poor air quality alert (1: issued; 0: not issued)**

You are interested in estimating the causal impact of the air-quality alert program on cardiovascular-related mortality. You propose to conduct a regression discontinuity (RD) analysis using data on AQI as your running variable and AQI=48 as your threshold. You define your exposure, the alert program, as a binary variable which equals 1 if an alert is issued and 0 otherwise. Your outcome is a daily count of cardiovascular-related mortality in the city between January 1, 2003 and December 31, 2012.

6) Is your study an example of a sharp RD or a fuzzy RD? Briefly explain your answer.

Solution

This is a sharp RD, as the probability of treatment (i.e., an alert being issued) is 0 below the threshold and 1 above it.

7) How would you state the research question you would be answering with this design in your own words? (1 pt)

Solution

We would estimate the effect of the air quality alert to residents on cardiovascular-related mortality when AQI=48.

8) Which of the following causal effects can be estimated by your RD design? (Select all that apply. Note: some answers are equivalent. Assume that in this setting, you cannot reasonably assume homogeneity of treatment effects along the running variable.) (2 pts)

- The average treatment effect (ATE)
- The average treatment effect (ATE) at the threshold
- The local average treatment (LATE, or the ATE among the compliers) at the threshold
- The average causal effect (ACE)
- The complier average causal effect (CACE)

Solution

All of these effects can be estimated, except for the ATE. In a sharp RDD, ACE = ATE at the threshold = CACE = LATE at the threshold.

9) Would it be plausible to assume homogeneity of treatment effects along the running variable? (1 pt)

Solution

The running variable is air pollution, and increasing air pollution will presumably lead to a greater number of cardiovascular related deaths. So the treatment effects are not likely to be homogeneous along the running variable.

10) State in your own words what the ACE means in the context of this study design? What do you need to assume for this ACE to be an internally valid causal estimate? Explain what these assumptions mean in the context of this study. (3 pts)

Solutions

The ACE in this study design would be the effect of the air pollution warning on cardiovascular mortality at the AQI threshold of 48. We'd need to assume continuous potential outcomes at the threshold. In practice, this means that AQI is not manipulated in order to cause air quality alerts to not be issued when they otherwise would be (or to be issued when they otherwise would not be), or that the threshold coincides or causes changes in other covariates that might affect cardiovascular related mortality.

As you're about to conduct the analysis, you find out that your prior understanding was incorrect, and that there is an additional criterion for issuing air-quality alerts. A Ministry official tells you alerts are occasionally not issued even though AQI is above the threshold, in order to avoid alert fatigue. Furthermore, alerts are occasionally issued when AQI is below the threshold when a team of advisors believes that other factors warrant an alert to be issued.

11) Does the new information you received in any way change the type of RD analysis you can do? Explain your answer.

Solution

Yes, now we are estimating a fuzzy RDD, as the probability no longer goes from 0 to 1 in crossing the threshold.

12) Would you still be estimating the ACE? Explain your answer. (1 pt)

Solution

Yes. Now we are identifying the CACE, rather than the ACE, as we now have a fuzzy RDD.

13) Does the new information change the assumptions needed to identify the causal estimand of interest? If so, explain the new assumption(s) needed to identify the causal effect of interest in the context of this study. (2 sentences)

Solution

Yes. In addition to our previous assumption of continuous potential outcomes, we now need to assume that there are no defiers at the threshold (monotonicity). In the context of this study, it means that the threshold increases the likelihood of an alert being issued across all days, such that no day would have received an alert only if AQI were below the threshold and would not have received the same alert had that day's AQI been above it.

Part 4: Conducting a regression discontinuity analysis - Kangaroo Mother Care

We will now use R to conduct a regression discontinuity design study. For these questions, you will be asked to write your own code to run RRD analyses and create plots. To help guide you, we have provided tips in the next section, but we encourage you to reach out to the teaching team and to others for feedback.

Close to 15 million babies worldwide are born preterm, i.e., at a gestational age of less than 37 weeks. One of the easiest and most promising interventions to increase the chance of children's survival is Kangaroo Mother Care (KMC). The primary feature of this method is skin-to-skin contact between the mother and the child with the objectives to keep the child warm and to encourage early bonding between mothers and their children.

While there are a few trials suggesting that KMC has beneficial effects on child survival, relatively little is known about the effect of this intervention on child development.

To further examine this effect, you have obtained clinical data on a sample of children delivered at a hospital in Rwanda (where KMC has been promoted for a few years), as well as data on children's socio-emotional development at age 2. The socio-emotional development variable (SEZ) is a normalized developmental score, with mean zero, and standard deviation of 1 (i.e. a z-score). The data include the following information for each child: gestational length in weeks ($gestationalAge$), reported directly by the clinic, the child's sex assigned at birth ($female$: 1 = female, 0 = male), and an indicator for whether or not the child received KMC (kmc: 1 = received KMC, 0 otherwise).

A colleague has informed you that most hospitals in Rwanda are supposed to strongly encourage mothers to use KMC for any pre-term delivery, but that KMC is less frequently encouraged when babies are not pre-term. Check whether this claim is true empirically. To do this, generate an indicator for preterm birth

using the $ifelse()$ command (1 if gestational length < 37 weeks, 0 otherwise) and plot a scatter of the probability of using KMC as a function of gestational length on both sides of the preterm indicator that you generated. (Tips on how to visualize this data are provided in the attachment.)

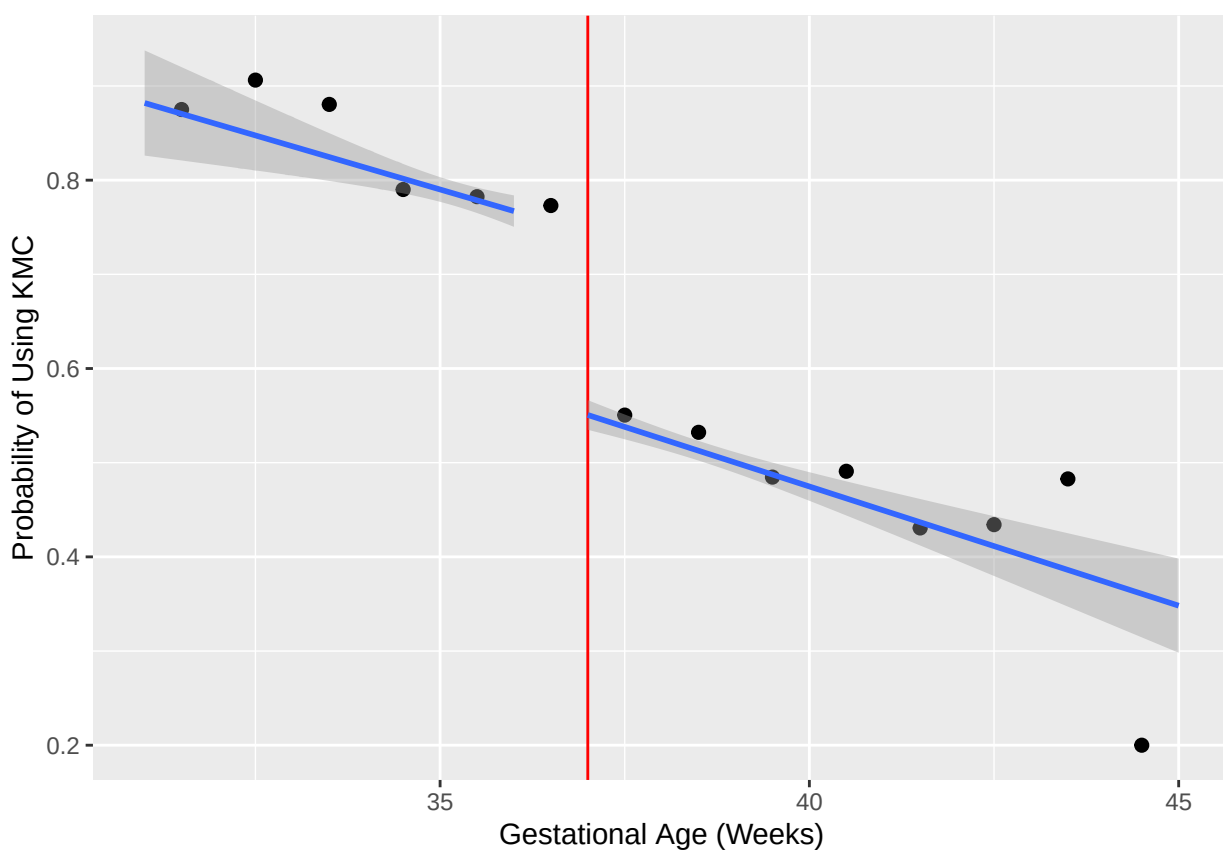
14) Load the data, create an indicator variable for pre-term birth, and graph a scatter plot with gestational age on the x-axis and pre-term birth on the y, grouped by pre-term designation. Paste your graph below. From your graph, what do you observe regarding the likelihood of KMC above and below the threshold, and what do you conclude about the type of discontinuity you are analyzing? (1 pts)

```
# Loading necessary libraries
library("sandwich")
library("lmtest")
library("tidyverse")
library("fixest")

# Loading data set
ps7.data <- read.csv("kmc.csv")     # Change here if you have renamed the file

# Creating a preterm indicator
ps7.data$preterm <- c(ifelse(ps7.data$gestationalAge<37, 1, 0))

# Graphing a scatter plot with linear time trends on each side of the preterm discontinuity
ggplot(ps7.data, aes(x=gestationalAge, y=kmc, group=preterm)) +
stat_summary_bin(fun='mean', bins=14, size=2, geom='point') +
geom_vline(xintercept = 37, color = "red") +
geom_smooth(method='lm',formula=y~x) +
labs(x = "Gestational Age (Weeks)", y = "Probability of Using KMC")
```



Solution

We can see that we have a fuzzy discontinuity, as the probability of receiving KMC is about 20 percentage points higher just under the gestational length threshold of 37 weeks vs. just above it.

15) Check whether the difference in the probability of doing KMC is statistically different for preterm babies by regressing KMC on sex, gestational length and the preterm indicator. (Please use a linear probability model with robust standard errors for this step). What coefficient do you get for the preterm indicator, and how do you interpret it? What does the F-statistic suggest about the strength of preterm status as an instrument for KMC? (1 pt)

```
partD <-  lm(formula = kmc ~ preterm + female + gestationalAge, data = ps7.data)
summary(partD)
```

```
##
## Call:
## lm(formula = kmc ~ preterm + female + gestationalAge, data = ps7.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.8839 -0.5062  0.2157  0.4689  0.6304
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     1.464846   0.127617  11.478  < 2e-16 ***
## preterm         0.190746   0.014394  13.252  < 2e-16 ***
## female          0.012147   0.008433   1.440     0.15
## gestationalAge -0.024893   0.003306  -7.529 5.48e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.473 on 12583 degrees of freedom
## Multiple R-squared:  0.07111,    Adjusted R-squared:  0.07089
## F-statistic: 321.1 on 3 and 12583 DF,  p-value: < 2.2e-16
```

```
coeftest(partD, vcov = vcovHC(partD, "HC1"))
```

```
##
## t test of coefficients:
##
##                  Estimate Std. Error t value  Pr(>|t|)
## (Intercept)     1.4648462  0.1286087 11.3899 < 2.2e-16 ***
## preterm         0.1907462  0.0142209 13.4131 < 2.2e-16 ***
## female          0.0121473  0.0084329  1.4405    0.1498
## gestationalAge -0.0248927  0.0033312 -7.4726 8.379e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Solution

The coefficient on preterm $\beta_1 = 0.1907462$ with a p-value $< .001$. This indicates that, holding all covariates constant, that preterm babies are 19 percentage points more likely to have received KMC than non-preterm babies.

16) One of the key tests to make sure there is no misreporting of the running variable (*gestationalAge*) is to look at local distributions. Create a centered variable for *gestationalAge* that is centered around the threshold, and then plot a histogram with 14 bins of *gestationalAge* including a line representing the

preterm cutoff. Paste your graph below and evaluate whether there is any visual evidence of bunching? (1 pt)

```
ps7.data$gestationalAge_centered = ps7.data$gestationalAge - 37
hist(ps7.data$gestationalAge_centered,
     main="Histogram for Gestational Age",
     xlab="Gestational Age above vs. below 37 weeks",
     col="gray",
     las=1,
     breaks=14,
     prob=TRUE)
  abline(v=0, col="red")
```

Solution

The distribution appears smooth across the cutoff indicating that there is likely not any systematic manipulation.

17) Explain one plausible scenario under which bunching of this data could occur. Do you think this scenario is likely? Why or why not? (2 pts)

Solution

It might be tempting to manipulate an assignment variable (in this case, gestational age) if it will allow someone to obtain a treatment or gain access to a program that they value. In this case, because anyone can do KMC (and it does not cost money) it is unlikely that the assignment variable is manipulated. Nevertheless, manipulation could occur if mothers lie about the start date of their pregnancy deliberately or if medical staff misreport gestational age in order to recommend KMC to mothers. Both situations seem rather unlikely.

18) For the rest of our analysis, trim the data to a bandwidth of 5 weeks above and below the threshold. Run a linear regression of *SEZ* on the *preterm*, *gestationalAge_centered*, and *female* variables on your trimmed data and using robust standard errors. How do you interpret your coefficient on preterm? (2 pts)

```
partI <-   feols(SEZ ~ gestationalAge + female + preterm, # Our covariates and main predictor
              data = subset(ps7.data, gestationalAge_centered > -5 & gestationalAge_centered <  5), # T
              se = "hetero") # Heteroskedasticity robust SE

summary(partI)
```

```
## OLS estimation, Dep. Var.: SEZ
## Observations: 12,219
## Standard-errors: Heteroskedasticity-robust
##                 Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)     0.105619   0.309863  0.340856 7.3322e-01
## gestationalAge -0.004650   0.008055 -0.577275 5.6376e-01
## female          0.091775   0.018058  5.082341 3.7832e-07 ***
## preterm         0.068994   0.032265  2.138388 3.2505e-02 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.997729   Adj. R2: 0.003364
```

Solution

The coefficient on preterm $\beta_1 = 0.068994$ with a p-value $= 0.0325$. This indicates that preterm babies have, on average, SEZ score which is 0.07 standard deviations higher than full term babies conditional on covariates among babies around the 37 week gestational age threshold.

19) Is the preterm coefficient you just estimated the complier average causal effect (CACE)? If so, explain what the CACE means in the context of this study. If not, explain what this preterm coefficient is estimating (e.g., ATE, ITT effect, LATE, etc.) (2 pts)

Solution

This estimate is not the CACE. Rather, this estimate is more akin to an intention-to-treat (ITT) estimate in that it regresses the outcome on the instrument directly. To get the CACE, we would need to scale this estimate by the probability of compliance, which we can do using the standard Two Stage Least Squares (2SLS) estimator.

20) Now conduct two-stage least squares estimation using preterm status as an instrument for kmc, and estimate the Complier Average Causal Effect (CACE), again using the trimmed data and robust standard errors. Interpret your coefficient. How does the coefficient compare to the estimate in part (20)? Explain why this is different. (1 pt)

```
cace <- feols(SEZ ~ gestationalAge + female # Our covariates
              | kmc ~ preterm, # endogenous variable ~ instrument
              data = subset(ps7.data, gestationalAge_centered > -5 & gestationalAge_centered <  5), # T
              se = "hetero") # Heteroskedasticity robust SE
summary(cace)
```

```
## TSLS estimation - Dep. Var.: SEZ
##                   Endo.    : kmc
##                   Instr.   : preterm
## Second stage: Dep. Var.: SEZ
## Observations: 12,219
## Standard-errors: Heteroskedasticity-robust
##                 Estimate Std. Error   t value   Pr(>|t|)
```

```
## (Intercept)    -0.440822   0.510130 -0.864136 3.8753e-01
## fit_kmc         0.366306   0.162424  2.255250 2.4135e-02 *
## gestationalAge  0.004741   0.011244  0.421658 6.7328e-01
## female          0.086551   0.017329  4.994581 5.9778e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.947626   Adj. R2: 0.100946
## F-test (1st stage), kmc: stat = 152.7    , p < 2.2e-16 , on 1 and 12,215 DoF.
##              Wu-Hausman: stat =   6.63282, p = 0.010023, on 1 and 12,214 DoF.
```

Solution

$\beta_{IV} = 0.366306$ with p-value $= 0.02413$ Assuming the identifying RD LATE assumptions hold, this estimate reflects the CACE, i.e., the effect of KMC on socio-emotional z-score at 2 years among compliers at the cut-off. This is larger than the estimate in (20), since we've scaled our estimate in (20) by the probability of compliance at the cut-off which is a value between 0 and 1.

21) Suppose you learned that parents of preterm infants were targeted with encouragement to come to additional follow-up care appointments for their infant and were provided with materials surrounding supporting their child's development. How would this change the way that you interpret what your RDD model is estimating? (2 pt)

Solution

We would no longer be estimating the LATE for KMC. If the KMC intervention was part of this package of encouragements delivered at the same time, and the indicator for kmc in the dataset is also the indicator for receiving this whole suite of interventions, then we would be estimating the LATE for the combined intervention. If they were delivered separately, and our probability of compliance is only for KMC and not for the other interventions, then our LATE estimate for KMC would be biased because we would have violation of the continuity of potential outcomes assumptions. Instead, we would then interpret the ITT we estimated in Q18 to evaluate the effect of a preterm designatioin on SEZ at the threshold.